# Energy Demand Projections for Urban Household Cooking in Ethiopia from 2022-2050: a MAED-Based Approach

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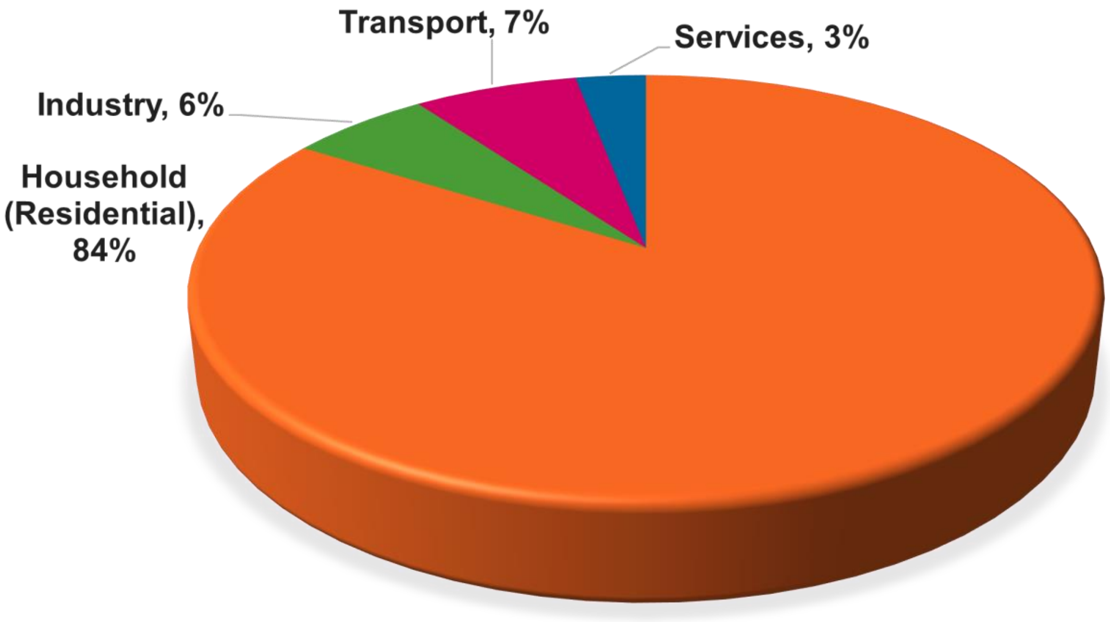
- Key Policy Insights, Conclusion and Future Works

# Context & Challenges



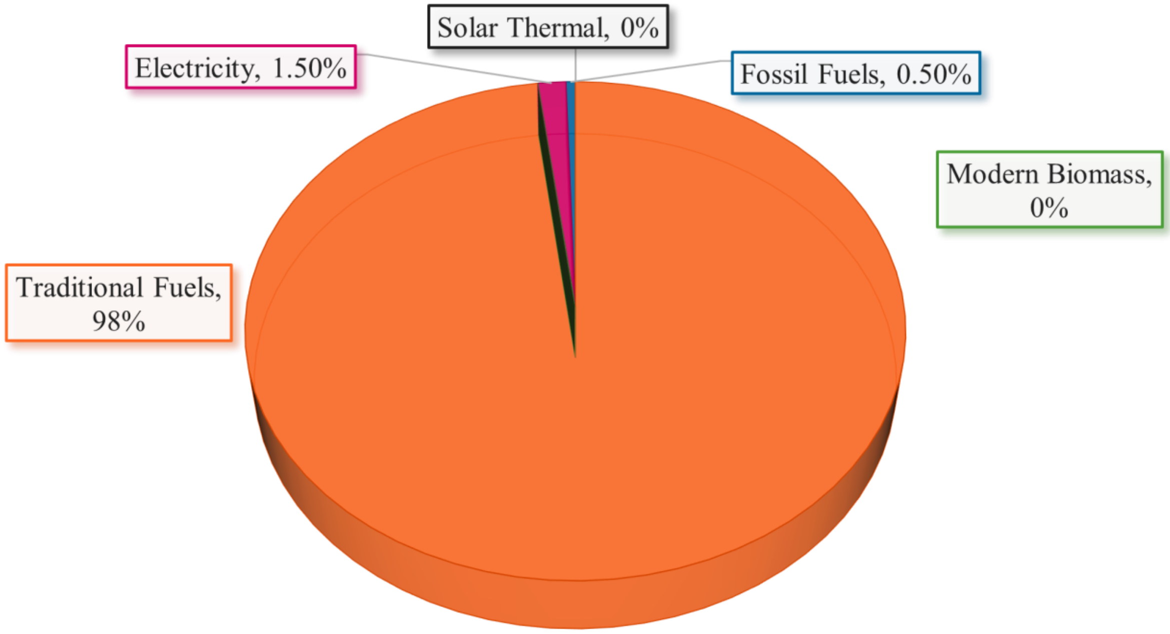
- 27 million urban population with 5.6 million households @2022 and rapidly growth at 2050.

Sectorial Share of Final Energy Consumption @2022




**Source:** Asfaw et al., 2024

Share of Energy Consumption for Cooking in The Urban Household Sector By Energy Forms




**Source:** AFRIC

# Research Questions

## ❑ Research Question

1. What will be the impact of **traditional fuel while** shifting to **clean cooking** and **adopting modern biomass technology** in Ethiopia household sector energy demand with reduction of carbon emission by 2050 ?

# Scenarios

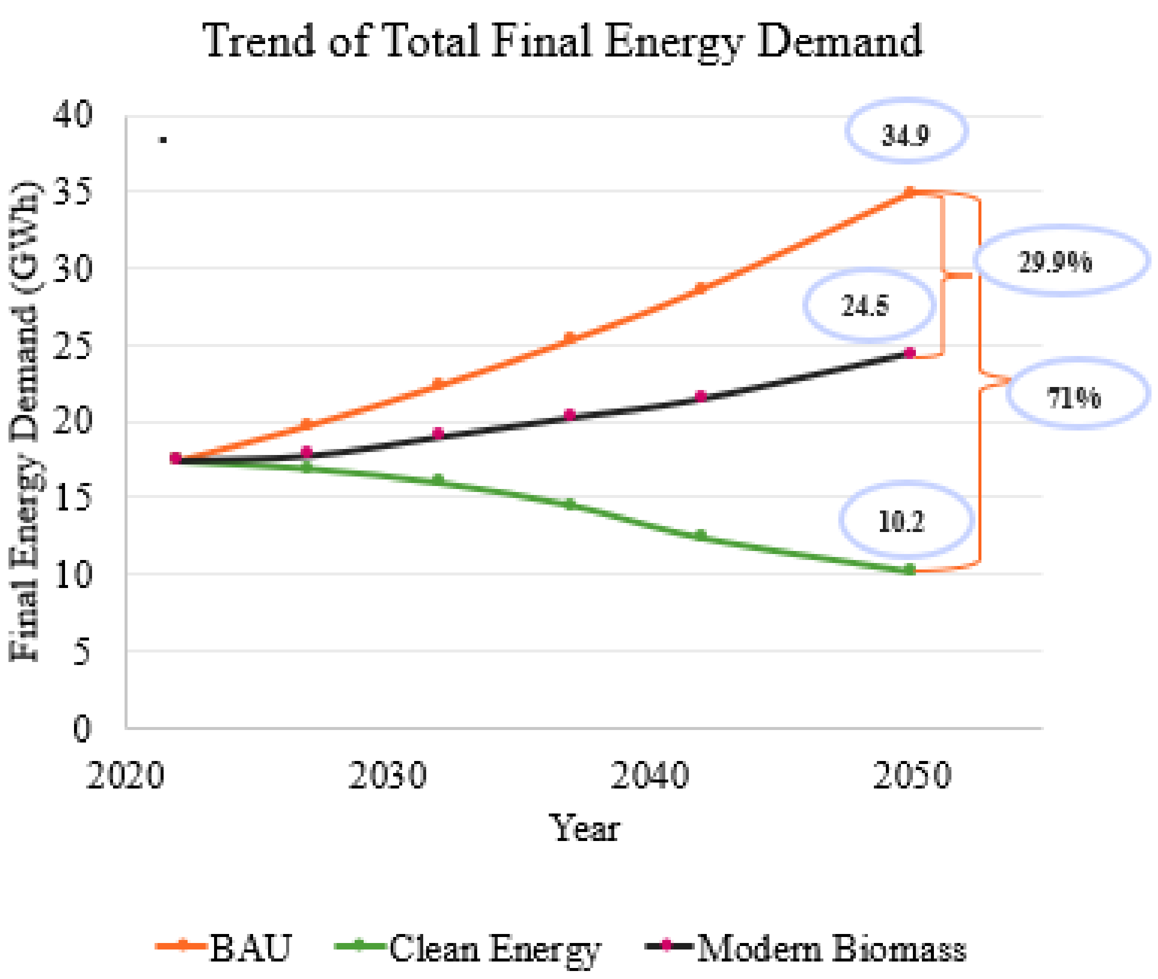


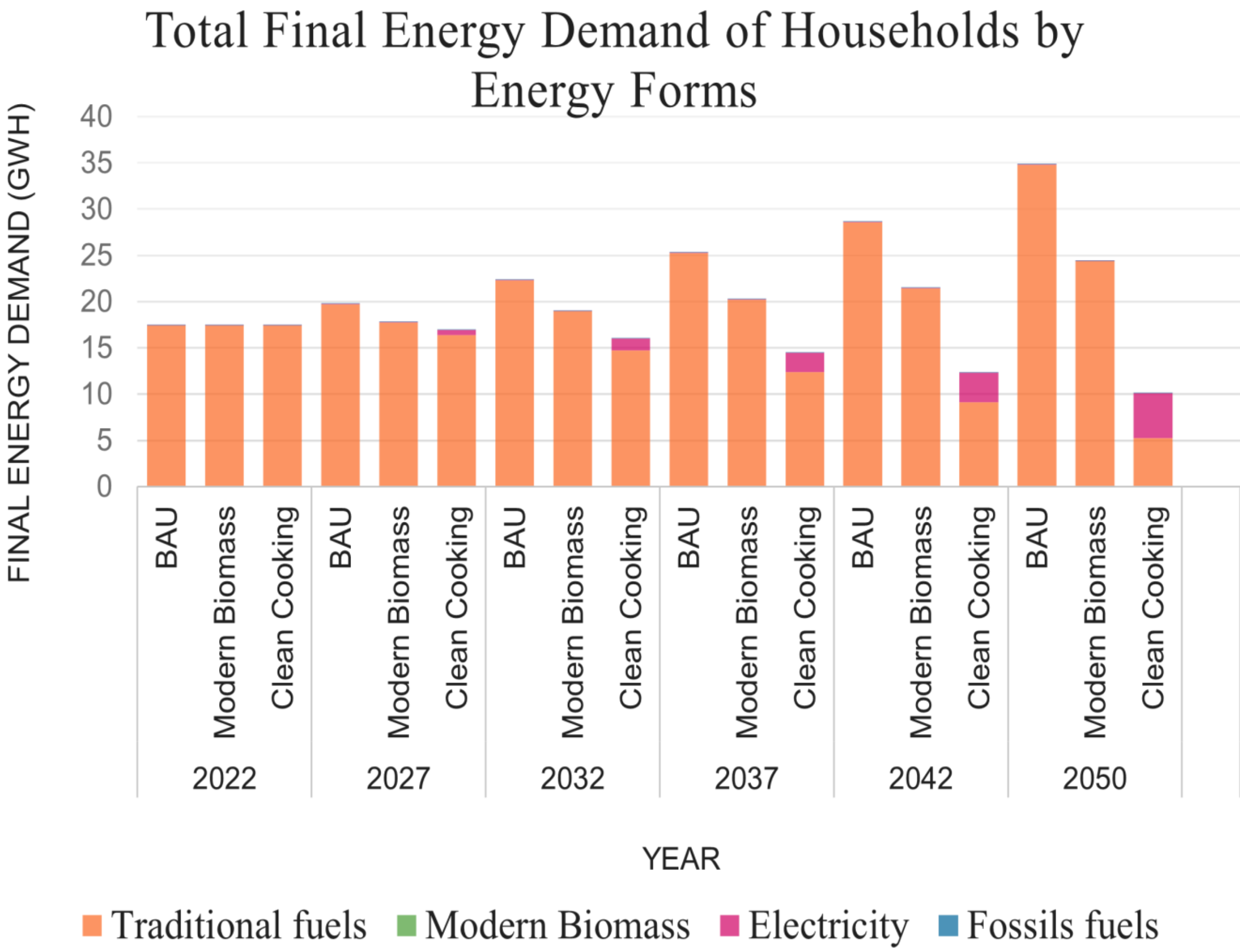
Using the Model for Analysis of Energy (MAED), three scenarios were investigated:

| Scenario Label | Scenario Description | Key Assumptions |
|---|---|---|
| BAU | Current trends continue | • Traditional fuel-dominated cooking (98%); slow electrification |
| Modern Biomass | Technology efficiency improvement | • Shift to adoption of modern biomass technologies by 30% @2050 |
| Clean cooking | Rapid electrification of cooking | • 85% electric cooking by minimizing traditional fuel @2050 |

# Results




Trend of Total Final Energy Demand
Final Energy Demand (GWh)
40
35
30
25
20
15
10
5
0
2020
2030
2040
2050
Year
34.9
29.9%
24.5
71%
10.2
BAU
Clean Energy
Modern Biomass


Total Final Energy Demand of Households by Energy Forms
FINAL ENERGY DEMAND (GWH)
40
35
30
25
20
15
10
5
0
BAU
Modern Biomass
Clean Cooking
2022
2027
2032
2037
2042
2050
YEAR
Traditional fuels
Modern Biomass
Electricity
Fossils fuels

# Results



**Reduction of CO2 emission**

BAU → Modern biomass: 6.16 million tons of CO2 reduced to 4.31 million tons of CO2 (29.2%)

BAU → Clean cooking: 6.16 million tons of CO2 reduced to 1.714 million tons of CO2 (70.8%)

# Key Policy Insights, Conclusion & Future Work

## Key Policy Insights and Actors

- Private sectors and development partner should subsidize efficient modern biomass technologies and prioritize efficiency standards in urban households.
- The minister of water and energy (MWoE) and utilities should adopt clean cooking policy from existing access to utilization by creating awareness for end users.

# Key Policy Insights, Conclusion & Future Work

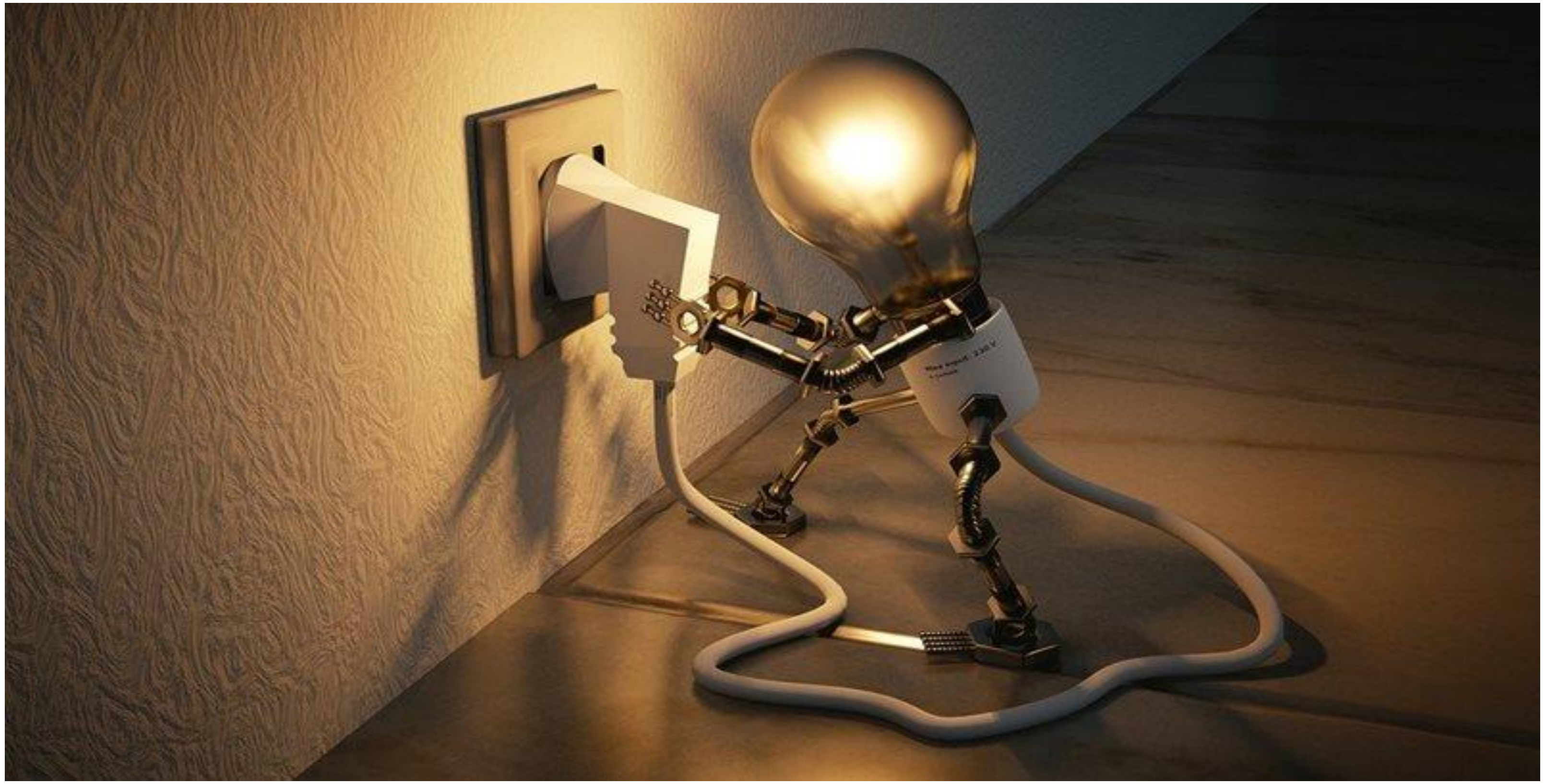
## Conclusion

- Urban household energy demand varies strongly by scenario.
- BAU nearly doubles energy demand by 2050
- Clean cooking and modern biomass reduce energy demand up to 71% &29.9%.
- Clean cooking: 6.16 million tons of $CO_2$ reduced to 1.714 million tons of $CO_2$

## Future Work

- Link demand results with power system expansion planning.
- Expand to additional end uses (appliances, lighting).

# Unplug Me, Save Energy, Save Life



## Thank You, Questions & Comments...